

Student Performance at Portuguese Schools

Project Summary & Problem Framing

The goal of this project is to analyze students data with an intention to create a machine learning model to predict a student's performance for a large Portuguese school system. The results of this study will be used to identify students who need assistance and intervention to improve their grades. The data set contains 395 rows and 35 columns including three types of variables: numerical (student's age, grades, etc.), ordinal (parent's education - from primary to high, etc.) and categorical (access to the internet, parent's job type, etc.).

The final grade (*G3*) is the target variable, the value of which would be predicted based on other independent variables (features). The selected numerical features are the student's age, a number of failures during past classes, student's absences, first (*G1*) and second (*G2*) term grades. There is a strong correlation between *G3* and *G1*(*G2*). The two separate cases were studied: when *G1* and *G2* are included in a model, and when they are not. Even though *G1* and *G2* are strong predictors that would significantly improve a model, leaving those variables out allows a model to be independent of these grades and predict a student success even before a school year starts. Despite a weak correlation between age, failures and student absences with the target variable, there are totally just a few numerical features in the data set, thus, a decision was made to include all of them in the study at this stage.

The selected categorical features are student's sex (*sex*), whether there is internet access at home (*internet*), and whether there is planning for higher education (*higher*). These are binary variables with two categories 'yes' or 'no'. The selected ordinal features are home to school travel time (*traveltime*), weekly study time (*studytime*) and how much a student goes out with friends (*goout*). The data contains a lot more interesting categorical and ordinal features, however, these six were selected at this stage, as the target variable is clearly distinct between various categories for these variables. For example, *G3* values are significantly different for various travel times. It does not seem reasonable to include features that do not have any effect on the target, however, this may be a point of further investigation.

Three regression models were used to predict student's performance: Linear Regression, Support Vector Machines (SVM) Regression and Lasso Regression. The model result scores were used to evaluate how good each model is, compare them and choose the best one. According to different metrics scores (RMSE, R-squared), Linear Regression works best for the case with *G1* and *G2* in the model including all the other features, explaining 77% of variability in the final grade target variable. Model priority is in favor of SVM Regression, however, if *G1* and *G2* are excluded from the model. The model reliability for the latter case is small. Even after fine-tuning the SVM parameter space, the model explains only 16 % of the target variable.

At this stage, none of the three models used are recommended for implementing into production. The next step would be experimenting with other features of the data set and using Ensemble

Learning methods, when predictions of a group of predictors are aggregated to achieve a better result. Collecting more data would definitely help better understand correlations between the variables, feature selection resulting in more precise model fits.

Data Overview

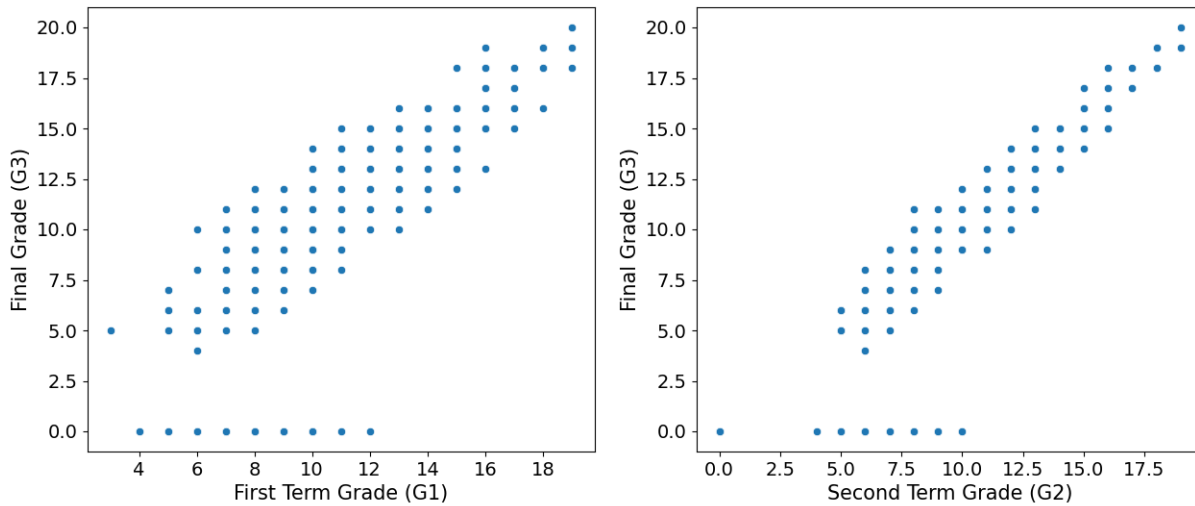
The student data table contains 35 columns and 395 rows. Out of 35 variables, 8 are numeric: *age* (age of a student), *failures* (number of past class failures), *absences_G1*, *absences_G2*, *absences_G3* (number of absences for previous G1, G2, G3 terms, respectively), *G1*, *G2*, and *G3* (G1, G2 terms and final grades, respectively). The rest are ordinal and categorical variables. Ordinal variables are encoded with integer numbers (0, 1, 2, 3, 4, etc.) with the order that has a specific meaning. For example, for *Medu* or *Fedu* (mother's or father's education, respectively), 0 means no education, 1-3 - primary to secondary education, and 4 - education higher than secondary. Examples of binary categorical variable can be *schoolsup* (extra educational support), *activities* (extra-curricular activities), *internet* (internet access at home) and so on, with categories *yes* or *no*., or multi class variables, e.g. *guardian* (student's guardian: 'mother', 'father', 'other'), *Mjob* (mother's job: 'teacher', 'health care related', 'civil services', 'at home', 'other') and so on.

The final grade (G3) is selected to be the *target* with the purpose to model and predict student's performance based on other features. The main goal of this study is to identify students who might need help at the beginning and during the school year to improve their grades.

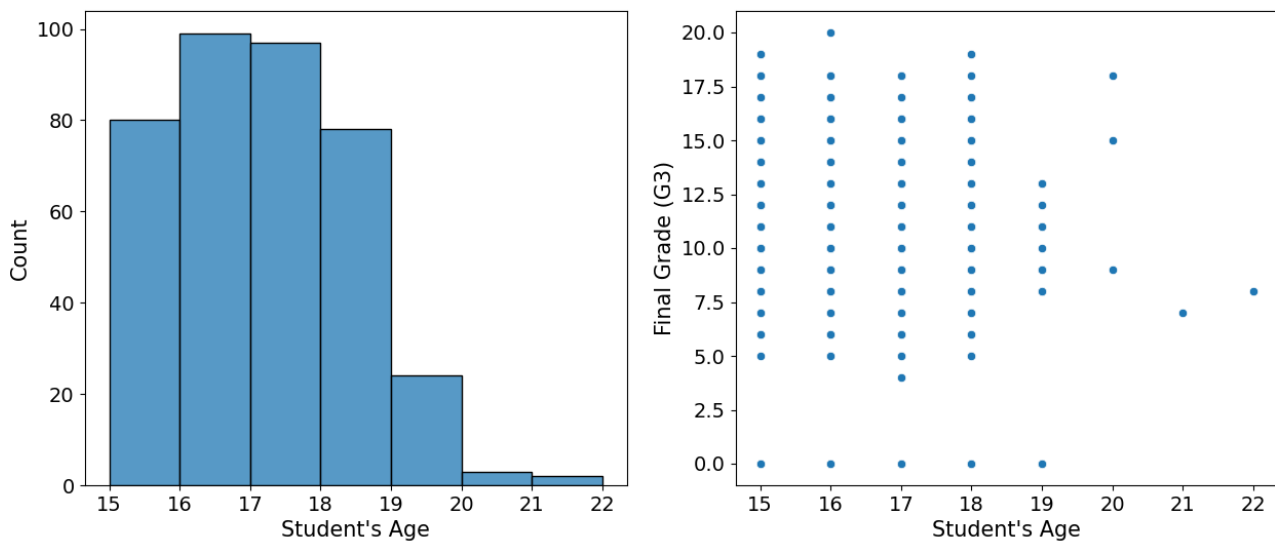
There is a small fraction (<4% per column) of missing values in the numeric columns: *age*, *absences_G1*, *absences_G2*, *absences_G3*. There are outliers for the target variable - final grade (G3): 38 instances with 0 grade, which seem more like NULL values than an actual grade. However, the decision was made to keep the outliers in the data, as there is no insight regarding the meaning of 0 grade within the Portuguese school system. There is clearly a majority of students in the 15-19 year old range, while the fraction of 20-22 year old students is substantially lower (<2%).

A Pearson correlation coefficient was calculated for all the numerical variables. A strong correlation was found for G1 vs. G3 and for G2 vs. G3 variables with the coefficients equaled to 0.8 and 0.9 (with 1 being a perfect correlation), respectively.

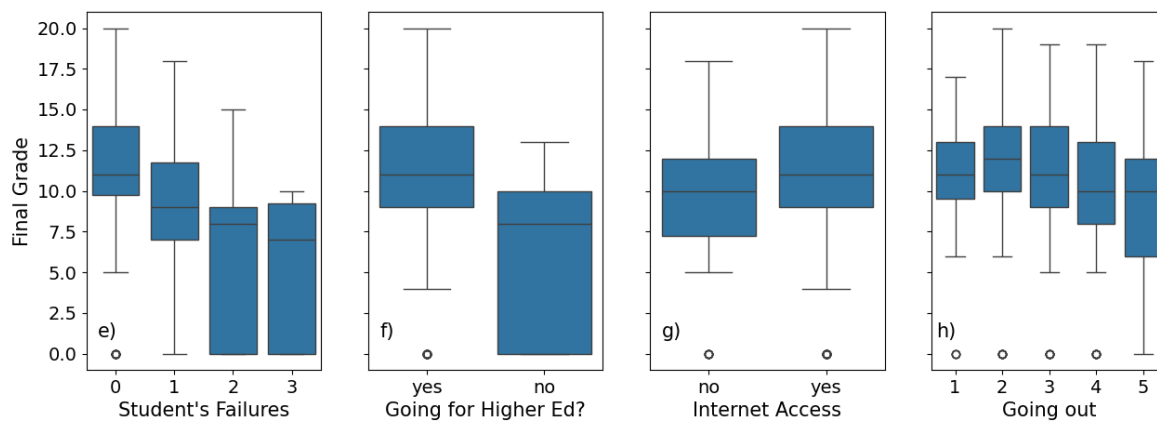
It makes sense that with more failures, the final grade is lower, which can be demonstrated by the computed negative correlation coefficient of -0.36. It is counterintuitive, however, to see no dependence of G3 on *absences_G1*, *absences_G2* or *absences_G3* (correlation coefficients are all lower than 0.05). The expectation would be to have a lower final grade with a higher number of absences.



It is important to include grades for the first and second terms (G1 and G2) in the model due to



their strong correlation with the target (G3). However, it is also important to be able to make predictions in advance of a school year using other important features, to intervene early and help a student to be successful. Thus, the two separate models are considered: with and without G1/G2 grades.



Analytical Insight

There is a student's age imbalance in the data set: majority of the students' age is between 15 and 19 years old (Fig. 1). The scatter graph shows no correlation between age and the final grade varying between 4 and 20 for the youngest group (15-19 year old) and between 7 and 18 for the 20-22 year olds (Fig. 2).

Fig. 1. Histogram of student age variation.

Fig. 2. Final grade vs. student's age.

Both G1 and G2 strongly correlate with G3 (Fig. 3): if the grade is high during any term, the final grade is also most likely high, and vice versa.

Fig. 3. Scatterplots of the final grade (G3) vs. the first (G1) and second (G2) term grades.

Fig. 4. Box plots of final grade (G3) distribution by various categories.

The final grade separated by categories for all categorical and ordinal variables was plotted on a series of box plots, some of which are presented at Fig. 4. Males are slightly more successful than females (Fig 3a). This may not be always the case, especially when more data is collected, but that's what it is in the current data set. Also, students at urban address have a higher final grade than those with rural address, which is shown by both their median and total ranges of values (Fig. 3b). G3 median value falls as travel time increases (Fig. 3c): students on average study better if they live closer to their school. Apparently, with greater study time (*studytime*) G3 increases (Fig. 3d), and on the contrary, more *failures* result in lower final grades (Fig. 3e). Students willing to pursue *higher* education would be more serious at studies, which is shown by their greater final grades (Fig. 3f). Students tend to study slightly better when they have access to the internet (Fig. 3g). One would expect students of better health to perform better, but surprisingly there is no clear correlation of health status with the final grade. Going out with friends (*goout*) seems to help to a better performance. However, the final grade starts to fall as going out count goes higher and higher (Fig. 3h).

Selecting numerical features. The following numerical variables were included in the models: *age*, *failures*, *absences_G1*, *absences_G2*, *absences_G3*, *G1*, and *G2*.

Absences_G1/G2/G3. One would expect that increasing number of absences would cause the final grade to decrease. However, there is a very weak correlation between these variables for a possible reason that those absences were recorded in the previous year and may not have an effect on a current year grade. On the other hand, one can observe that fewer absences are still associated with the highest final grade, whereas with many absences, students never reach the top grade category. Thus, it is still important to keep the class absences variables in the model as they may provide an additional constraint. For simplicity, the absences from all the three terms were added up and used as a single feature.

G1/G2. Considering these variables would greatly improve the model due to their strong correlation with G3, it is important though to see the model performance without them and consider other features that may effect the final grade to predict a student's performance early in a year. The result would effect one's decision to intervene early with a student's learning process and provide a useful advise.

Age. It is interesting to see the effect of a student's *age* on a final grade. Even though the correlation with the target is very weak, and there are only a few students of older age (>19 years old), high grades of the latter are rare, which is an important constraint.

Failures. There is clearly a good negative correlation of student's failures with the final grade: G3 falls with more failures. Thus, *failures* would be an important predictor.

Categorical features. The following categorical variables were selected for the model: *sex*, *higher*, *internet*.

Sex. A significant difference in student's performance for males and females is not expected. G3 values for males though plot slightly higher than for females, which is interesting but hard to explain. If additional accumulated data provide evidence that this relationship is statistically significant, student's sex would be an important contributor to the model.

Higher, *internet*. There is no surprise in how G3 is affected by these two features. Planning for higher education and having internet access at home yield higher grades. This obvious correlation is the reason to have them in the model.

Ordinal features. The following ordinal features were selected for the model: *studytime*, *traveltime*, *goout*.

Included in the model *studytime* and *traveltime* have four categories: 1, 2, 3, 4, with 1 meaning the least amount of time and 4 - the most. It is straightforward to suggest that with more study time, there will be a better grade, and on the opposite, longer travel time results in a worse grade. These features are expected to be reliable predictors and thus, would be a good contribution to the model.

Interesting variation of G3 for different *goout* categories (1-5), where the target variable slightly increases, as a number of outings rises from category 1 to 2 and then slightly falls as *goout* continues to rise. That means students still do well if they go out with friends sometimes, but apparently, going out too much (categories 3 and higher) may not be a good idea.

Methodology

Models. Three separate regression models were considered for this study: Linear Regression, Support Vector Machines (SVM) and Lasso Regression. We use the word *regression* as opposed

to classification, as we are trying to predict a numerical continuous target using both numerical and categorical features. *Linear Regression* makes a prediction through a linear equation that sums up selected features multiplied by their coefficients (weights). This algorithm computes those coefficients defining whether and how statistically significant (important) they are. *Lasso Regression* is one of the regularized versions of a linear regression, at which a regularization term is added to the regression equation. Increasing a regularization parameter (α) that controls regularization prevents a model from overfitting, allowing it to generalize better with new data. Important characteristic of Lasso regression is that it automatically eliminates weights (setting them to zero) of the least important features, leaving only the most important ones. *SVM* works best for small-sized data, which is appropriate for the current data set. SVM regression algorithm finds the best fit through data, aiming to fit as many data points as possible within a specific error margin controlled by parameter ϵ . Smaller ϵ regularizes the model: adding more training data points will not affect its predictions.

The data set was divided into a training set and a testing set. Rows for both sets were randomly selected with the proportion of 80% for the training set and 20% for the testing set. The training set was then used to explore, clean and transform the data into the shape suitable for predictive modeling. Missing values were imputed with the column median for numerical features and with the most frequent category for ordinal and categorical variables. Numerical features were scaled relative to each other to be on a similar scale. Categories of categorical variables were encoded with zeros and ones necessary for regression models. The above described models were then applied to fit the training set. The testing set was transformed and then used as new data to check the model performance.

Metrics. The model performances were evaluated using Root Mean Squared Error (RMSE) and R-squared. *RMSE* measures the distance between the fitted curve/plane and actual data values. The bigger the RMSE, the bigger that distance and, thus, the worse the fit. *R-squared* indicates what percentage of variance in the target variable is explained by independent variables. It varies between 0-1, with 1 when the fit is perfect with 100% variance explained, and 0 when none of the variance is explained.

Grid search. To achieve better results, a search for the best combination of SVM regression model parameters was performed using *grid search*. The selected parameters are C - cost parameter adjusting a trade off between better generalization and minimizing an error, and ϵ - sensitivity of a model to new data. Thus, the algorithm searches the best combination between C (0.01, 10, 100) and ϵ (0.001, 0.1, 0.5, 1) for the two cases: with and without G1 and G2 grades.

Key Results

As expected, all three models perform better when G1 and G2 features are included. In the case with G1 and G2 grades, RMSE values are 1.90, 2.55 and 2.17 for linear regression, SVM and

Lasso, respectively, and 4.3, 4.2 and 4.4 when G1 and G2 are not included. According to these metrics, linear regression is the best choice and SVM is the worst in case the first and the second term grades, G1 and G2, are included. Out of the three considered models, the SVM regression would work best when G1 and G2 are excluded.

Testing set. To check how well the models generalize with new data, their performance was measured with a testing set. RMSE values for linear regression are 2.18 and 4.32 including or excluding G1/G2 from model, respectively, which is close to the scores of fitting the training set. These corresponds to R-squared of 0.77 and 0.09, respectively. Fine-tuning the SVM model improves the fit to the data slightly, resulting in RMSE values of 2.27 and 4.15 with and without G1/G2, respectively, compared to the higher values of 2.55 and 4.20. R-squared values for SVM model are 0.75 and 0.16 with and without G1/G2, respectively. Testing the model demonstrates that both models generalize well with new data. Linear regression model still performs better than SVM even after fine-tuning.

According to the R-squared values the selected models explain 75-77 % and 9-16% of the final grade variations with and without G1/G2, respectively. Even the highest scores are still quite low, thus, the models using current features are not recommended for implementing. The models still need improvement possibly by adding additional features and trying more advanced models (*e.g.* Ensemble Learning) exploring a wider parameter space.

Every school is interested in student's success. The main purpose of this study is to predict a student's performance based on specific features but also to give specific recommendations for future successful performance based on data insights. Basic recommendations to students would be more study time, moderate going out frequency, thinking about and planning for higher education, choosing a school that is located close to home.

Conclusion

The data set for this study contains 395 rows and 35 columns most of which representing categorical and ordinal features. Many of these features, even those that seem like important predictors (family support, health, family relationship) do not have any correlation with the target, which makes it hard to select features for modeling. Interestingly, students with school support or family support, have lower grades on average than those who do not, which seems counterintuitive. Including or excluding G1 and G2 features reveals a big difference on a model performance. Having those features in the model makes it stronger. It would be more useful to be able to make predictions based on other features, not related to current year's success / failures, but other features have very weak correlation if any with the target making a model practically unreliable (very low scores).

The data set with 395 instances is too small to have reliable statistics. Thus, the next step would be to collect more observations and see if that improves models' scores. It is worths to pay more

attention to outliers and missing values; cleaning the data in this project was automated and not very thorough. Only three models were used in this study. Using more comprehensive models (*e.g.* Ensemble Learning) with optimizing usage of specific features (instead of manual selecting) and more thorough parameter grid search would result in a better model performance.